QA Skills Assessment

Manual Testing

1. List all the questions and clarifications that you need to plan and design the test cases for this feature
   i. What information should be displayed for the **main product**, such as name, seller, category, and price?
   ii. What conditions define a product as **'related'** to the main product? Is it determined only by category and price range, or are additional filters such as brand, material, ratings, or seller location also considered?
   iii. When considering the price range for related products, what exact range should be used?
   iv. What does **"same category"** mean? Does it refer only to the exact sub-category, or does it also include factors like brand, material, or color?
   v. How is a **"best seller"** defined for a related product? Is it based on the number of sales, ranking algorithm, or other criteria?
   vi. What should happen if there are **no related products** available for a main product?
   vii. If there are **more than 6 related products** matching the same category, price range, and best seller criteria, how should we decide which 6 products to display? What additional filters or prioritization should be applied?
   viii. How exactly should we **define a best seller** for a related product? What conditions need to be considered in the filtering process?
   ix. Does **'Similar Sponsored Items'** refer to the same products as **Best Sellers**?
   x. When products are listed in **different currencies**, how should we check that they fall within the same price range?
   xi. Does product quantity update **in real time**, and should the related products list reflect these updates automatically?
   xii. Should the related products section behave the same way for **logged-in users and guest users**?
   xiii. Can the **same product from different sellers** appear as two separate related items for the main product?
   xiv. What information needs to be displayed for the **main product** and for **related products** (e.g., name, price, image, seller, rating)?
   xv. How should the **order of products** be determined when searching by name?
   xvi. How should the **order of related products** be decided?
   xvii. Are there any differences in the related products feature between **web and mobile platforms**?

xviii. When a user clicks **"See All"** in the similar sponsored products section, which products should be listed and how should their order be determined?

xix. Can a product exist **without an image**? If not, how many images should each product have?

2. Create a test strategy document for this scenario

Test Strategy: eBay Main Product Related Products Feature

- **Project Overview**

This feature is about eBay shopper products validations. There's a Main product as per the search criteria, then there should be related products, Up to 6 related products need to be displayed and they are the best sellers of same category and price range. Here main goal is to validate whether Main Product is displaying with correct details and the related products are displaying correctly as per the requirement

- Functional Flow Validation
- Navigation Accuracy
- User Experience

- **Project Scope**

In-Scope

- Verify the Main product information with its related products
- Verify related products count
- Verify related product price range
- Verify related products order and display information
- Validate product link is working for all the related products
- UI Validations such as product images, see all link, category names

Out of Scope

- Add to cart and checkout flow
- Performance testing
- Database validations

- **Risk Analysis**

Risk	Risk Mitigation
Data fluctuation price, product quantity changes	Use mocked APIs use fixed test data for price validation

UI Locator changes	Regularly review and update locators, use table locators like IDs
Currency differences	Convert all prices to one currency before comparision
Timeout issues	Increase timeout util files in Playwright fw
Site unavailability	Rerun the tests using retry logic, notify team asap if site is unreachable

- **Types of Testing and Approach**

Automation Testing

Playwright to handle regression testing, critical path testing and cross browser testing

Postman to handle Manual API testing

Manual Testing

UI validation

Exploratory testing

- **Test Environment, Tools**

OS	Windows 11
Framework	Maven+TestNG
Automation Tool	Playwright for Java
IDE	IntelliJ IDEA
Language	Java 25.0.2
Web Browser	Chromium

- **Test Data**

Search Name	leather wallet for men
Main Product	First result returned from search
Prince Range	Main Product Price+/-100
Category	Lether, Men’s wallet
Search Procut Name	"leather wallet for men"
Max count for related Products	**6**
URL	https://www.ebay.com
Price Range	Main Product Price ± $100

- **Entry and Exit Criteria**

Entry Criteria	**Exit Criteria**
eBay website is accessible at https://www.ebay.com	All test cases have been executed at least once
Development of the Main Product and Related Products feature is complete	All the High Priority and High Severity (critical path) test cases are passed
Test environment is set up	No critical bug found
All test cases are reviewed and approved	All Unit Test are passed
Test data is prepared	Test objectives are completed
Relevant APIs/API Docs are available	Test reports shared with team members
Test runners are configured properly (TestNG.xml)	

- **Test Metrics**

Matric	**Formula**	**Target**
Automation Coverage	(Automated TCs / Total TCs) × 100	80%
No of Open defects	Total Open Defects	0 defects Critical/High
Test Cases Executed	Count of TCs run	100%
Defects Fixed	Total bugs resolved	100% of Critical/High
TC Pass Rate	(Passed / Executed) × 100	>80%
Defect Leakage	Production Bugs / Total Bugs × 100	0%

3. Based on the assumptions above, please prepare a full test suite

Detailed Test Cases-Positive Test Cases

TC No	Description	Precondition	Test Steps	Expected Results	Priority
TC001	Verify search is working for Products	User navigate to eBay website	1.Go to Home page 2.Add product name 3.click Search	List of products appear as per the entered name in search	high
TC002	Verify Product result	TC001	1.view Product results	Main Product results should be displayed with images and product information	high
TC003	Verify maximum 6 related products are displayed	1. User is on Wallet Main product detail page 2. Related products section is visible	1. Scroll to related products section 2. Count the number of items displayed	Maximum of 6 related products are displayed	High
TC004	Verify related products belong to same category as main product	1. User is on wallet product detail page 2. Main product category info is available	1. Scroll to related products section 2. Check category of each related product	All related products belong to wallet/leather category	High

TC005	Verify related products are within price range of main product	1. User is on wallet product detail page 2. Main product price is visible	1. Note main product price 2. Check price of each related product 3. Compare — should be within ± $100	All related product prices are within main product price ± $100	High
TC006	Verify each related product displays an image and heading	1. User is on wallet product detail page 2. Related products section is visible	1. Scroll to related products section 2. Check each product card for image 3.Check the heading and description of the product	Every related product card displays a product image	High
TC007	Verify clicking a related product navigates to correct product page	1. User is on wallet product detail page 2. Related products section is visible 3. At least 1 related product is shown	1. Scroll to related products section 2. Click on first related product card 3. Check new page URL	User is navigated to the clicked product detail page All the links of the related products are working	High

TC008	Verify related products section is visible	TC001	1.View Main product details 2.scroll down to 'Similar Sponsored Items'	Related products are visible	High
TC009					
TC010					

Negative Test Cases

TC No	Description	Precondition	Test Steps	Expected Results	Priority
TC011	Verify empty search shows all categories page	1. User is on eBay homepage 2. Search box is empty	1. Clear the search box 2.Click Search	Page navigates to all product categories	High
TC012	Verify no duplicate products in related list	1. User is on wallet product detail page 2. Related products section is visible	1. Scroll to related products section 2. Collect all related product titles 3. Compare titles for duplicates	All related product titles are unique	Medium
TC013	Verify out of stock items do not	1. User is on wallet	1. View each related product card	No out-of-stock items appear in	High

	appear in related products list	product detail page	2. Click each item and verify availability	the related products section	
TC014	Verify zero related products also visible	1.find a unique product 2.go to its detail page	1.View Main product 2.View related product section	Unique main products without related product are available	Medium

Edge Test Cases

TC015	Verify related products update when navigating to different wallet	1. User has viewed product A 2. User navigates back to search results	1.Open wallet A note related products 2. Go back to results 3. Open wallet B 4. Check related products	Related products section shows wallet B items and not wallet A items	High

Assumptions

- Both logged-in and guest users can see the related products section
- Price range is defined as Main Product Price ± $100
- Maximum of 6 related products is a hard limit
- Minimum of 1 related product must appear if section is visible
- Similar Sponsored Items is the same as Related Best Seller Products
- Best seller is determined by eBay's internal sales ranking algorithm
- If category and price is same then consider the alphabetical order of product name to be displayed

- Out of stock items are excluded from the related products list
- Main product itself should NOT appear in its own related products list
- All related product links should work without issue
- No duplicate products should appear in the related products section
- Auction items are excluded, only fixed-price listings are shown

4. Assume that you found five issues in this scenario and create a bug report for those five issues.

Bug Report 1

Field	Details
Bug ID	BUG-001
Title	More than 6 related products displayed in the section
Severity	High
Priority	High
Environment	Windows 11, Chrome, eBay.com
Reported By	ThiliniC
Status	Open
Precondition	User is on wallet product detail page
Steps to Reproduce	1. Search 'leather wallet for men' 2. Open first product 3. Scroll to related products section 4. Count items
Expected Result	Maximum 6 related products are displayed
Actual Result	8 related products are displayed — exceeds the maximum limit of 6
Attachment	screenshot_bug001.png

Bug Report 2

Field	Details
Bug ID	BUG-002
Title	Main product appears in its own related products list
Severity	High
Priority	High
Environment	Windows 11, Chrome, eBay.com
Reported By	ThiliniC
Status	Open
Precondition	User is on wallet product detail page
Steps to Reproduce	1. Open any Main product 2. Note the Product ID from URL 3. Scroll to related products section 4. Check all related product links for same item ID
Expected Result	Main product should NOT appear in its own related products list
Actual Result	Main product appears as 3rd item in related products section
Attachment	Evidences(screenshots/error logs)

Bug Report 3

Field	Details
Bug ID	BUG-003
Title	Related product from different category appears in wallet related list

Field	Details
Severity	High
Priority	High
Environment	Windows 11, Chrome, eBay.com
Reported By	ThiliniC
Status	Open
Precondition	User is on wallet product detail page
Steps to Reproduce	1. Open wallet product page 2. Scroll to related products section 3. Check category of each related product
Expected Result	All related products should belong to wallet/leather category
Actual Result	One related product is a 'Men's Belt' — belongs to a different category
Attachment	Evidences(screenshots/error logs)

Bug Report 4

Field	Details
Bug ID	BUG-004
Title	Related product price is outside the defined price range
Severity	High
Priority	High
Environment	Windows 11, Chrome, eBay.com
Reported By	ThiliniC
Status	Open

Field	Details
Precondition	User is on wallet product page priced at $25.00
Steps to Reproduce	1. Open wallet product priced at $25 2. Scroll to related products 3. Check price of each related product
Expected Result	All related product prices should be within $25 ± $100 range
Actual Result	One related product is priced at $250 — well outside the ± $100 range
Attachment	Evidences(screenshots/error logs)

Bug Report 5

Field	Details
Bug ID	BUG-006
Title	Out of stock products are showing in related products section
Severity	High
Priority	High
Environment	Windows 11, Chrome, eBay.com
Reported By	ThiliniC
Status	Open
Precondition	User is on wallet product detail page
Steps to Reproduce	1. Open any wallet product page 2. Scroll to related products section 3. Click on each related product 4. Check product availability
Expected Result	Only in-stock products should appear in the related products section
Actual Result	One related product shows "Out of Stock" label — unavailable item is displayed in the related products list

Field	Details
Attachment	Evidences(screenshots/error logs)